

## 6 Global Effects of Stretching

The process of science does not typically follow the amazing discoveries of Newton, Einstein, or Curie or our other historical luminaries. Even these amazing scientists had to build upon the prior work of others. Newton is reported to have stated, “If I have seen further, it is by standing on the shoulders of giants.” These incredible scientists while building upon the foundational work of others, subsequently provided giant leaps over their predecessors. Most researchers are scientific bricklayers and we add a few more bricks to the foundation of knowledge and hopefully a bit like a good architect suggest slightly new designs (research directions) to investigate.

### **Crossover or non-local muscle fatigue**

In the early 2000s a relatively new concept termed crossover fatigue was reported (36, 45). Surprisingly to many people, if you unilaterally fatigued one limb and then tested the same (homologous) contralateral muscle, impairments in muscle strength, power, or endurance could be experienced. I read those papers and was intrigued by the finding that one arm or leg muscle could produce less force or power and fatigue more quickly without even doing any physical activity. All you had to do was fatigue the opposite (contralateral) muscle. Naturally lacking any substantive imagination of my own, to come up with amazing new ideas, I borrowed their idea and found a research gap. The gap was that the early research typically examined the contralateral homologous muscle (unilaterally fatigue the quadriceps and test the contralateral quadriceps). Thus, my amazing revelation (sarcasm alert!) was to investigate if non-exercised contralateral or even ipsilateral (same side) heterologous muscles would be impaired by prior fatigue of another muscle. As it could be either contralateral or ipsilateral heterologous muscles, we thought crossover fatigue was not totally appropriate, so we coined the term “non-local muscle fatigue” (NLMF) as the impairments could be anywhere in the body (global effects) (21, 22, 23). With our first review of the literature in 2015 (22), we found that whilst the literature was conflicting (32 of 58 measures showed NLMF), there was evidence that NLMF effects were more apparent when the lower body (e.g., quadriceps) was tested, isometric and

cyclical fatigue interventions had a greater incidence and magnitude of NLMF, and fatiguing or extended duration testing protocols exhibited greater NLMF. Hence, we pursued these issues in subsequent research. Six years later, we published a meta-analysis of 52 studies (278 measures), which did not support the existence of a “general” NLMF effect (e.g., strength and power deficits); but there was evidence for endurance-based (i.e., time to task failure) NLMF impairments (6). However, another interesting gap arose from the “shoulders of giants” and we wondered whether there was crossover, non-local, or global effects of stretching. Could you stretch one muscle and have increased range of motion (ROM) or flexibility in other joints and muscles?

### **Global Stretching Effects on Range of Motion**

Lo and behold, there were acute global (non-local and crossover) effects of unilateral stretching. Some of the first studies demonstrated that if one hamstrings was stretched then the contralateral hamstrings (hip flexion ROM) experienced greater flexibility (16), if participants stretched their hip abductors (groin), they experienced greater shoulder (horizontal abduction) ROM and if they stretched their shoulders, then there was an improvement in hip flexion (hamstrings) ROM (8).

We also wondered about the possibility of cross education, stretch training effects. Cross education has been demonstrated since 1894 by Scripture (46) with only two participants (Miss Smith and Miss Brown) who showed improvements in contralateral muscular control and power after unilateral practice. Since then, there have been innumerable studies that show a cross transfer of training effects for motor skills, muscle strength, power, and other attributes in all segments of the population (18, 19, 24, 31). We compared single versus twice daily unilateral stretch training sessions on hip flexion ROM of the stretched and contralateral, non-stretched legs. Both one and two per day stretch training sessions over two weeks improved both active static (5.9 percent vs 5.2 percent ROM improvements for stretched and contralateral respectively) and ballistic (5.01 percent vs. 2.3 percent ROM improvements for stretched and contralateral respectively) ROM (12). Nakamura et al. (40) compared the cross-education effects of unilateral stretch training with four weeks of high- vs. low-intensity stretching on dorsiflexion ROM, muscle stiffness, and architecture. They found significant, moderate magnitude, contralateral dorsiflexion ROM increases with the high intensity stretch group but not the low intensity group. There were no changes in muscle stiffness or architecture.

You would not expect everybody to agree, would you? In contrast, a ten-week stretch training programme of the plantar flexors (three days/week, four repetitions of 30 seconds each with 30 seconds rest between stretches) did not induce cross transfer improvements in ROM, but did produce significant strength gains (41). Similarly, the Konrad group did not find global (non-local effects on heterologous muscles or regions) improvements in shoulder

extension ROM after seven weeks of static stretching and foam rolling on the sole of the foot (28) or with seven weeks of pectoralis muscle static stretching (both studies intervened three days/week with three exercises of five minutes each) on ankle dorsiflexion ROM (29).

You will notice in this second edition, that many meta-analyses are reported. During the COVID-19 pandemic, we and others could not collect data as all labs were closed due to the pandemic. In order to stay busy, stay relevant, and keep contributing to science, we published a number of meta-analyses. Meta-analyses are a great way to survey the literature and quantitatively evaluate what has worked and not worked and then highlight new research directions to pursue. One of these meta-analyses (4) showed that an acute bout of unilateral passive static stretching induced moderate magnitude improvements in passive ROM with non-local, non-stretched joints. Participant sex, trained state, stretching intervention intensity, and duration did not moderate these results. We also noticed that more than 240 seconds of stretching induced large magnitude increases in non-local ROM compared with moderate magnitude enhancements with shorter (< 120 seconds) durations of stretching. It was emphasized earlier in this text that static stretching should not exceed 60 seconds per muscle group as part of a warm-up, otherwise there could be subsequent performance impairments.

### **Global Stretching Effects on Performance**

But, as the early acute stretching research from our and other labs highlighted that prolonged static stretching without a full warm-up (addition of dynamic activities before and after static stretching) could induce performance impairments (7, 9, 15, 26), we wondered whether this was possible in a non-stretched, non-exercised muscle. As some research demonstrated declines in electromyography (EMG), EMG/M-wave (muscle action potential wave) ratios and voluntary activation (e.g., interpolated twitch technique, volitional wave), it was suggested that there may be a reduction in central (neural efferent) drive (49, 50). Furthermore, muscle stretching can diminish persistent inward currents (PIC) at the motoneurons due to reduced muscle spindle facilitation (51). PICs can increase the gain of the motoneuron output by amplifying the synaptic inputs to the motor unit facilitating force output (51). Would these deficits occur solely with the motoneurons of the stretched muscle or would they spread to contralateral motoneurons?

In addition, it has been postulated that long duration fatigue interventions and perhaps prolonged stretching may have physio-psychological implications, which could result in a mental energy deficit when experiencing stretching to the point of discomfort or pain. This mental fatigue, could reduce the focus, attention, and concentration, needed to achieve maximal or near maximal neuromuscular activation (22), especially with prolonged or repeated contractions. Steele (47) discusses the “perception of effort” which is defined as the individual’s perception of the required effort to accomplish a

specific task or set of tasks,. It is possible that the prolonged stretching to the point of discomfort may have elevated the perception of effort involved in the subsequent task, which could contribute to global or non-local performance impairments.

A systematic review of the limited literature investigating the global effects of prolonged static stretching (mean stretching characteristics:  $6.3 \pm 2$  repetitions of  $36.3 \pm 7.4$  seconds with  $19.3 \pm 5.7$  seconds of recovery between stretches) revealed that both the stretched and contralateral, non-stretched, muscles revealed small magnitude force deficits (5). However, the power of these findings was not strong as the frequency of studies with these effects were similar with three measures demonstrating deficits, and four measures showing trivial changes.

### **Mechanisms Underlying Global Stretching Effects**

Non-local or global ROM improvements could be attributed to increased stretch or pain tolerance (32, 33, 35, 48). Increased stretch tolerance proposes that the musculotendinous unit can tolerate greater stress without a change in tension for a given length (33, 34). Augmented stretch tolerance is often promoted as an underlying mechanism for the increase in the ROM of the stretched muscle (7, 10, 15). We unilaterally treated the hamstrings with a menthol based topical analgesic to decrease pain sensitivity and discovered an increased passive static and ballistic hip flexion ROM in both the treated and contralateral limbs (52). Whether the increased pain (stretch) tolerance is primarily psychological or physiological (e.g., diffuse noxious inhibitory control theory suggests a release of endorphins and enkephalins in response to pain), the effects would be felt throughout the body (global) and not just directed to the affected (stretched) region, limb, or joint.

Other possibilities may involve a downregulation of sympathetic excitatory nerve stimulation (4). The goal of pain perception is to alert the physiological system to potential threats or damage and therefore upregulate the sympathetic nervous system (fight or flight) (38). However, with prolonged stretching near or at the point of discomfort, the perception that stretching would be an injury threat would diminish and therefore the fight or flight response (increased sympathetic excitation) would be downregulated. This would result in muscle relaxation permitting greater extensibility. Once again, sympathetic stimulation or parasympathetic relaxation is a global response and all muscles would experience this effect and not just the stretched muscles.

Further muscle relaxation may be achieved with sustained static stretching (e.g., 30 seconds or more) by attenuating motoneuron spindle reflexive (excitatory) activity (3, 20). As early as 1959 (42), stretch-induced changes in the myotatic reflex and inverse myotatic reflex were associated with reciprocal modifications in contralateral motoneurons of cat hind limb muscles. Type I and II muscle spindle afferents not only innervate spinal motoneurons (44) but also connect to the somatosensory and the primary motor cortex (43). As

the signals project to the cortex, the decrease in reflex derived excitability would have a global, full body reach. The decreased sympathetic nervous system stimulation and disfacilitation of excitatory spindle afferents (neural inhibition) could potentially increase non-local ROM while also potentially impairing performance. However, a study by Hadjizadeh Anvar et al. (2) reported contralateral ROM improvements with no significant spinal (H-reflex) and corticospinal excitability (motor evoked potentials with transcranial magnetic stimulation) changes suggesting that afferent excitability of the spinal motoneurons and corticospinal excitability may not play a role in global muscle's ROM increases. Contradicting the Hadjizadeh Anvar findings, Masugi et al. (37) reported that plantar flexors' muscle stretching (dorsiflexion stretch) demonstrated inhibition of monosynaptic spinal reflexes, for both the stretched and non-stretched muscles of the ipsilateral leg. More work is needed in this area to determine possible neural effects.

It has also been suggested that myofascial chains may play a role (30, 53, 54). The concept of myofascial chains posits that there is a continuous transmission of tension with the myofascia, which covers and extends from lower to upper body muscles (30). This continuity can either contribute to force transmission or if the myofascia becomes more extensible or compliant in one area, it can release tension in other areas of the body. For example, lower limb stretching provided increases in cervical ROM and was attributed to myofascial chains (13, 54). Wilke and colleagues (53) suggested that tissue strain and stress can be transferred from one body region to another, accentuating global connectivity. However, Wilke et al. (54) did not directly measure changes in fascial structural and thus stretch tolerance or reduced sympathetic excitation might have been major contributors. As the force transfer with myofascial chains or meridians predominately align longitudinally (11, 17), obliquely (39) as well as transversal to synergists and antagonists (25), these myofascial chains may not contribute substantially to an increased extensibility of a contralateral homologous muscle (i.e., quadriceps to quadriceps). Furthermore, the aforementioned two Konrad studies (28, 29) did not reveal a non-local increase in ROM when the stretching occurred either at the shoulders (test ankle dorsiflexion ROM) or the sole of the foot (test shoulder extension ROM). So, although we have seen improvements when stretching the hamstrings positively affecting the shoulders and vice versa, if there is an effect of myofascial chains it may not be able to significantly transfer effects over this greater cephalocaudal distance.

### **Global Effects of Foam Rolling**

Similar to the crossover or global effects of stretching, foam rolling has also demonstrated increased contralateral ROM following an acute (single session) bout of ipsilateral (unilateral) foam rolling. The evidence is not as comprehensive with chronic foam rolling training effects (27). Once again, similar to global effects of stretching, reduction in pain perception is suggested as the

most probable mechanism for contralateral ROM increases (27). As further evidence of global decreases in pain perception, unilateral rolling of the plantar flexors provided similar increases in pain pressure threshold (less pain sensitivity) in one study that monitored calf muscle tender points (1) and another study that examined the degree of pain with high frequency evoked tetanic contractions (14). In both cases, without even touching the affected muscle, all we had to do was roll the contralateral muscle and pain was reduced. With lower pain sensitivity you can push yourself through a greater ROM with less discomfort.

In terms of global or contralateral performance effects, the Konrad et al. scoping review (27) reported no changes in muscle performance (i.e., maximum voluntary isometric contraction, maximum voluntary dynamic contraction) after an acute unilateral foam rolling bout (just 4 studies with seven measures). However, one of the four studies found substantial (large magnitude) deficits in rate of force development of the contralateral limb following unilateral foam rolling of the ipsilateral leg (55). Ye et al. (55) postulated that the impairment might be related to a decrease in motor unit recruitment. The only two studies evaluating chronic unilateral foam rolling training found moderate magnitude enhancements. With the small number of studies in this scoping review Professor Konrad warned that their findings should be interpreted with caution.

## Summary

Stretching a particular muscle or joint can have global ramifications for the body with increases in flexibility or ROM not only with the stretched muscle and joint but other non-stretched muscles. This may be attributed to a global increase in stretch (pain) tolerance, attenuated sympathetic stimulation or perhaps decreased reflex excitation. Non-local or global stretching effects would be important for training and rehabilitation, when a musculotendinous injury prevents training or activity. However, the finding of greatest non-local passive ROM with more than four minutes of stretching should not be prescribed for warm-ups but for distinct stretch training sessions. There is evidence that performance impairments associated with prolonged static stretching can also be transferred to contralateral muscles. Thus, unilateral stretching could maintain or induce greater contralateral flexibility without exacerbating the injury to the affected muscles or tendons.

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